PES UNIVERSITY, BANGALORE
Department of Computer Science and Engineering

Software Requirements Specification

Smart Complaint Management System
Version 1.0
Prepared: 06-10-2026
Team Members: Dhawal Pathak — PES1UG24CS151
GANESH M — PES1UG24CS168
Harshitha P — PES1UG24CS186

Revision History

Name	Date	Reason For Changes	Version
<Team member>	06-10-2026	Initial SRS for mini-project	1.0

1. Introduction

1.1 Purpose

This SRS specifies the requirements for the Smart Complaint Management System, a web-based platform for lodging, tracking and resolving complaints. It defines scope, users, interfaces, functional/non-functional requirements, security requirements and traceability.

1.2 Intended Audience

Development team, project owner, testers, faculty evaluators and maintainers.

1.3 Product Scope

The system centralizes complaint submission and administration. A registered user can create a complaint with title, description and category, view submitted complaints and track status. An administrator can view/filter complaints, update status and record resolution information.

1.4 References

- SE Mini-Project Deliverables Part-1, PES University, 2026.
- Supplied PES University SRS template.
- Supplied PES University SAD template.
- Supplied PES University Test Plan template.

1.5 Overview

Section 2 describes the product and users. Section 3 specifies interfaces, functional/non-functional requirements, security, quality attributes, business rules and domain requirements. Appendices provide a glossary and requirement traceability matrix.

2. Overall Description

2.1 Product Perspective

The proposed system is a new self-contained web application. The logical flow is User → Web UI → Application/API → Database, with an Administrator using the same application through role-based access.

2.2 Product Functions

- Registration/login/logout
- Complaint creation and validation
- View and track own complaints
- Administrator login and role-based access
- Administrator complaint listing/filtering
- Status updates and resolution remarks
- Persistent user and complaint records
- Clear error feedback

2.3 User Classes and Characteristics

User Class	Characteristics	Main Functions
Registered User	Basic web literacy; authenticated account.	Register/login, submit complaints, view own complaints, track status.
Administrator	Authorized staff with elevated	View/manage complaints, update

User Class	Characteristics	Main Functions
	privileges.	status, record resolution.

2.4 Operating Environment

Modern desktop/mobile browser, backend application server and persistent database. The exact backend framework and database should be finalized by the team.

2.5 Design and Implementation Constraints

- Only authenticated users may submit complaints.
- Administrator functions are role restricted.
- Status changes follow the supported workflow.
- Secrets must not be hard-coded or committed.
- Implementation must remain practical for the mini-project environment.

2.6 Assumptions and Dependencies

- Users have browser/network access.
- Database is available when the application runs.
- At least one administrator account is provisioned.
- Deployment and third-party services may be finalized during implementation.

3. Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interface

- Login/registration clearly shows validation errors.
- Complaint form has labelled mandatory fields.
- User dashboard displays complaint status/details.
- Admin dashboard provides listing, filtering and management controls.

3.1.2 Software Interfaces

- Frontend communicates with backend through HTTP/HTTPS APIs.
- Backend communicates with the selected database.
- Authentication state uses the selected secure session/token mechanism.

3.1.3 Communications Interfaces

- HTTP/HTTPS for client-server communication.
- HTTPS should be used in deployment.

3.2 Functional Requirements

ID	Feature	Requirement	Priority
FR-01	User Registration	Allow a new user to create an account using required information.	High
FR-02	User Authentication	Authenticate registered users before protected functions are available.	High
FR-03	Complaint Submission	Allow an authenticated user to submit a complaint with title, description and category.	High
FR-04	Complaint Validation	Reject submissions when mandatory fields are missing or invalid.	High
FR-05	View Own Complaints	Allow a user to view	High

ID	Feature	Requirement	Priority
		complaints submitted by that user.	
FR-06	Track Complaint	Display current status and details of a user's complaint.	High
FR-07	Admin Access Control	Restrict administrative functions to authenticated administrator accounts.	High
FR-08	Admin Complaint Management	Allow administrators to view and filter complaint records.	High
FR-09	Status Update	Allow administrators to update complaint status using the supported workflow.	High
FR-10	Resolution Details	Allow administrators to record resolution remarks.	Medium
FR-11	Data Persistence	Persist users and complaints so records remain after a session ends.	High
FR-12	Error Feedback	Provide clear error messages for failed operations or invalid input.	Medium

3.3 Non-Functional Requirements

ID	Attribute	Measurable Requirement
NFR-01	Performance	For normal load, 95% of user-facing API requests should respond within 2 seconds.
NFR-02	Availability	Deployed system should achieve at least 99% availability during evaluation, excluding scheduled maintenance.
NFR-03	Security	Passwords shall use secure one-way hashing and never be stored in plaintext.
NFR-04	Authorization	Protected operations shall verify the user's role before privileged actions.
NFR-05	Confidentiality	Credentials and complaint data shall use HTTPS in deployment.
NFR-06	Usability	A first-time user should locate complaint submission without administrator assistance.
NFR-07	Reliability	A failed database/API operation shall not create a misleading successful complaint state.
NFR-08	Maintainability	Frontend, backend and data-access responsibilities shall be modular.
NFR-09	Scalability	Design should permit additional users/categories/admins without redesigning the workflow.
NFR-10	Testability	Functional requirements shall be testable using repeatable test cases.

3.4 Security Requirements

ID	Requirement
SEC-01	Authenticate users before protected access.
SEC-02	Enforce role-based authorization for administrator operations.

ID	Requirement
SEC-03	Do not store or return passwords in plaintext.
SEC-04	Validate and sanitize user-supplied complaint fields.
SEC-05	Do not expose sensitive account information in authentication errors.
SEC-06	Keep database credentials and token keys outside committed source code.

3.5 Software Quality Attributes

- Correctness: ownership/status data remains consistent.
- Usability: forms and statuses are understandable.
- Maintainability: modules have clear responsibilities.
- Reliability: failures do not silently lose data.
- Security: authentication, authorization and secure credential handling are mandatory.

3.6 Business and Domain Rules

- A complaint belongs to the authenticated user who submitted it.
- Only an administrator can change administrative status.
- Resolved complaints should contain resolution information.
- Users only see their own complaints unless an explicit public/administrative view exists.

Appendix A: Glossary

Term	Meaning
Complaint	User-submitted issue requiring attention.
Administrator	Authorized user who manages complaints.
Status	Current processing stage such as Pending, In Progress or Resolved.
RBAC	Role-Based Access Control.
API	Application Programming Interface.
SRS	Software Requirements Specification.

Appendix B: Requirement Traceability Matrix

Requirement ID	Description	Architecture/Design Reference	Test Case
FR-01	User Registration	Authentication	TC-01
FR-02	User Authentication	Authentication	TC-02, TC-03
FR-03	Complaint Submission	Complaint Service	TC-04
FR-04	Complaint Validation	Complaint Service	TC-05
FR-05/06	View and Track Complaint	Complaint Service	TC-06
FR-07	Admin Access Control	Authorization	TC-07
FR-08	Admin Complaint Management	Admin Service	TC-08
FR-09	Status Update	Admin/Complaint	TC-09
FR-10	Resolution Details	Admin/Complaint	TC-10
NFR-01	Response Time	Application	TC-11
NFR-03/04	Security/Authorization	Security	TC-07, TC-12